

STAGE 2: LOCAL WEB UI - QUICK START GUIDE

What You Built

A local web application that:

- ✓ Processes bank statements with one click
 - ✓ Shows visual spending dashboard (pie chart, bar chart)
 - ✓ Lets you review and fix uncategorized transactions
 - ✓ Saves processed files for month-over-month comparison
-

Your Project Structure

```
expense_tracker/
├── app.py          # Streamlit web app (NEW - Stage 2)
├── expense_categorizer.py  # Backend logic (Stage 1)
├── requirements.txt    # Python dependencies (NEW)
├── data/             # Saved processed files (auto-created)
│   └── processed_statements/
└── README.md         # Documentation
```

HOW TO RUN - Step by Step

Step 1: Install Streamlit and Plotly

Open VS Code terminal (`Ctrl + ``) and run:

```
bash

pip install -r requirements.txt --break-system-packages
```

Note: The `--break-system-packages` flag is needed on your system (Windows 11 with Python 3.12+)

What this installs:

- `streamlit` - Web framework for the UI
- `plotly` - Interactive charts (pie chart, bar chart)
- Keeps existing: `pandas`, `openpyxl`

Wait for: "Successfully installed streamlit-X.X.X plotly-X.X.X..."

Step 2: Run the Web App

In the same terminal:

```
bash  
  
streamlit run app.py
```

What happens:

1. Terminal shows: "You can now view your Streamlit app in your browser"
2. Browser opens automatically at `http://localhost:8501`
3. You see the Expense Tracker web interface

If browser doesn't open: Copy the URL from terminal and paste in browser

Step 3: Use the App

A. Upload Your Bank Statement

1. Click "Browse files" button
2. Select your Excel file (`bank_transaction_2025.xlsx`)
3. Wait for "✅ Processed XXX transactions"

B. View Dashboard (Tab 1: Dashboard)

- See total spent, transaction count, uncategorized count
- Pie chart shows spending by category
- Bar chart shows top categories
- Category breakdown table

C. Review All Transactions (Tab 2: All Transactions)

- Filter by category dropdown
- Filter by date range
- Search by description
- View full transaction table

D. Fix Uncategorized (Tab 3: Fix Uncategorized)

- Shows only Uncategorized and Miscellaneous
- Each transaction shows full details:
 - Date, Amount, Type, Full description
- Select correct category from dropdown
- Click "✓ Update" to save change

E. Save Your Work

1. Go back to Dashboard tab
 2. Click " Save Processed File" button
 3. File saved to `data/processed_statements/expenses_YYYYMMDD_HHMMSS.xlsx`
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Step 4: Stop the App

When you're done:

In VS Code terminal: Press `Ctrl + C`

Confirmation: Terminal shows "Stopping..." and returns to prompt

Keyboard Shortcuts

While app is running:

- `Ctrl + C` - Stop the app
- `R` in browser - Refresh/reload app
- `Ctrl + Shift + R` - Hard refresh (clear cache)

VS Code shortcuts (from guide):

- `Ctrl + `` - Open/close terminal
 - `Ctrl + S` - Save files
 - `Ctrl + P` - Quick file search
-

What the Interface Looks Like

Dashboard Tab

Expense Tracker

Upload Bank Statement (Excel)

[Choose file]

bank_transaction_2025.xlsx

Processed 1,192 transactions

Dashboard

All Transactions

Fix

Spending Overview

Total Spent

₹788,015

Transactions

1,192

Uncategorized

126

Avg

₹661

Pie Chart

(Categories)

Bar Chart

(Top Categories)

Category Breakdown Table

Category	Total	Count	Percentage
Food	₹316,000	478	40.1%
Misc	₹258,000	391	32.8%

Save Processed File

Fix Uncategorized Tab

Fix Uncategorized Transactions

 126 transactions need review

Show: ☒ Uncategorized only ☐ Miscellaneous ☐ Both

Date: 15-Jan-2025

Category

Amount: ₹1,299.00

[Dropdown ▼]

Type: UPI

[✓ Update]

Details: UPIOUT/500555797628/amazon@...

Current Category: Uncategorized

Date: 18-Jan-2025

Category

Amount: ₹2,450.00

[Dropdown ▼]

Type: UPI

[✓ Update]

Details: UPIOUT/503717213037/goindigo...

Current Category: Uncategorized

Where Files Are Saved

When you click "Save Processed File":

Location: `data/processed_statements/`

File name format: `expenses_20250212_143025.xlsx`

- Format: `expenses_YYYYMMDD_HHMMSS.xlsx`
- Example: February 12, 2025 at 2:30:25 PM

What's saved:

- All transactions with categories
- Any manual edits you made
- Ready for Excel, further analysis, or comparison

Troubleshooting

"Module not found: streamlit"

```
bash
```

```
pip install streamlit plotly --break-system-packages
```

App doesn't open in browser

1. Look for URL in terminal: `http://localhost:8501`
2. Copy and paste in browser manually

"Address already in use" error

Another Streamlit app is running:

1. Find the terminal running it
2. Press `Ctrl + C` to stop
3. Run `streamlit run app.py` again

Changes not showing

Hard refresh: `Ctrl + Shift + R` in browser

Upload button not working

1. Make sure file is `.xlsx` format
2. File should have header at row 10 (standard bank format)
3. Check terminal for error messages

Workflow Example

Monthly Bank Statement Processing:

1. **Download** bank statement from your bank
2. **Open** Expense Tracker (`streamlit run app.py`)
3. **Upload** the Excel file
4. **Review** Dashboard - see spending breakdown
5. **Fix** uncategorized items (Tab 3)
6. **Save** processed file with timestamp
7. **Compare** with previous months (files saved in `/data`)

Time: ~5-10 minutes per month

What Each Tab Does

Tab 1: Dashboard

Purpose: Quick visual overview of spending

When to use:

- First look after uploading statement
- Monthly review of spending patterns
- Before/after fixing uncategorized items

What you see:

- Summary cards (totals, averages)
 - Pie chart (category distribution)
 - Bar chart (top spenders)
 - Full category breakdown table
-

Tab 2: All Transactions

Purpose: Detailed transaction review and search

When to use:

- Find specific transaction
- Review all transactions in a category
- Filter by date range
- Verify categorization accuracy

Features:

- Filter by category dropdown
 - Date range picker
 - Search box (searches description)
 - Full sortable table
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Tab 3: Fix Uncategorized

Purpose: Categorize items that need review

When to use:

- After initial upload (fix uncategorized)
- Review miscellaneous items
- Improve categorization accuracy

What you do:

1. Read full transaction details
2. Choose correct category from dropdown
3. Click "✓ Update"
4. Repeat for all items
5. Go to Dashboard and Save

Accuracy-first design:

- Shows complete transaction info
 - Full particulars/description visible
 - Date, amount, type all displayed
 - Make informed decision before categorizing
-

Next Steps

Stage 2 is COMPLETE when:

- ☒ App runs in browser
- ☒ You can upload and process statements
- ☒ Dashboard shows charts correctly
- ☒ You can categorize uncategorized items
- ☒ You can save processed files

Ready for Stage 3?

Stage 3 = Cloud Deployment (Optional)

- Host on Streamlit Cloud (free)
- Access from anywhere
- Share with family (if needed)
- No local setup required

Not needed if:

- You're happy running locally
 - Desktop-only usage is fine
 - No need for remote access
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What You Learned

Stage 2 introduced:

- Streamlit framework for web apps
- Plotly for interactive charts
- Session state management
- File upload/download in web apps
- Tab-based navigation
- Clean, minimal UI design

Stage 1 + Stage 2 together:

- Backend logic (Python) + Frontend UI (Streamlit)
 - Separation of concerns (app.py uses expense_categorizer.py)
 - Local data persistence
 - Professional-looking web application
-

Tips for Best Results

Categorization Workflow

1. Upload statement
2. Check Dashboard - note uncategorized count
3. Go to "Fix Uncategorized" tab
4. Start from top, work down
5. Save after every 10-20 edits (just in case)
6. Final save when done

File Organization

- Keep one folder for all months
- Save with descriptive names if needed
- Delete test files to keep it clean
- Back up saved files monthly

Performance

- Close browser tabs you're not using
 - Restart app if it gets slow ((**Ctrl+C**), then re-run)
 - Don't upload huge files (>5000 transactions at once)
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? Questions to Ask Yourself

Is Stage 2 working?

- ☐ App opens in browser
- ☐ Upload works
- ☐ Charts display correctly
- ☐ Can categorize transactions
- ☐ Can save files

Ready for Stage 3?

- ☐ Do you need cloud access?
- ☐ Want to access from phone?
- ☐ Share with others?

If all Stage 2 checkboxes ☒ = **Stage 2 COMPLETE!**

If any Stage 3 checkboxes ☒ = Time for Stage 3

Otherwise = **You're done! Use it monthly.**

Current Status: STAGE 2 COMPLETE ☒

Next: Test the app, then decide Stage 3 or stop here